A 5-nm 60-GS/s 7b 64-Way Time Interleaved Partial Loop Unrolled SAR ADC Achieving 35.2dB SNDR up to 32 GHz

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***Abstract*— A 60-GS/s 7b 64-way time interleaved (TI) analog-to-digital converter (ADC) with analog front end (AFE) is described. The presented converter features a non-binary partial loop unrolled (LU) SAR SubADC architecture that leverages multiple comparators, thus enabling better tradeoff between noise and power compared to conventional SAR. Offsets mismatches among comparators of each SubADC are calibrated in background by detecting patterns in the SAR output decisions. This results in no need for any analog hardware reconfigurability or additional phase overhead. Fabricated in 5-nm technology, the prototype AFE and ADC deliver 35.5 and 35.2dB signal to noise and distortion ratio (SNDR) till 20 and 32 GHz, respectively, and draw 109.3 mW from 0.9 V supply.**

***Index Terms*— Optical communication, partial loop unrolled (LU), SAR, serializer/deserializer (SerDes), time interleaved (TI) analog-to-digital converter (ADC).**

I. Introduction

OPTICAL communications are a key enabler for artificial intelligence (AI) and accelerated computing. In the next-generation datacenters optimized for AI training, several AI servers are interconnected using high-speed and low-latency links that for 100-Gbps speed and beyond are often implemented using small form factor pluggable optical modules. Due to their small size, these modules have tight power constraints that result in a strong push to optimize the power of both the optical and the electrical components. Among them, the CMOS die, which houses the serializer/deserializer (SerDes) transceivers and performs all the equalization and retimer functionalities, is a key contributor.

For these applications, the analog-to-digital converter (ADC)-based SerDes receiver (RX) mostly use time interleaved (TI) ADCs to achieve low-power and high-speed operation [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. In this performance range, ADC input capacitance is an important parameter as it is driven by the analog front end (AFE) and may contribute to limit the overall bandwidth of the system. To minimize this capacitance, flat interleaving a relatively small number of high-speed SubADC channels is a possible option, which is also attractive from an ADC power efficiency perspective [1], [11]. In fact, flat interleaving does not require multiple sampling operations, hence reducing the amount of kT/C noise. However, this approach is still limited when trying to push sampling rate close to the maximum achievable with the available technology.

For wireline transceivers where this is indeed the case, hierarchical time interleaving is usually adopted [3], [4], [5], [6], [7], [8]. In this scheme, a reduced number of track and hold (TH) units integrated in the ADC front end sample the input signal at the maximum speed allowed by the technology. The sampled information is then distributed among many SubADC using TH buffers. SubADC speed and resulting interleaving factors are instead optimized to be amenable to DSP processing, thus avoiding power overhead in the digital domain. This speed range is often close to optimum for SubADC power efficiency as well. However, in hierarchical time interleaving, the input signal is typically sampled at least twice, hence degrading the overall noise budget. Furthermore, the power contributions of the ADC front end and TH buffers have to be considered in the overall converter power performance.

In this context, it is thus of paramount importance to optimize the noise-power efficiency of the SubADC, which in most cases uses SAR-based architectures.

In this article, we describe a 60-GS/s 7b 64-way TI ADC + AFE, which employs a non-binary partial loop unrolled (LU) SAR SubADC architecture that features better noise-power tradeoff compared to a conventional SAR architecture. Comparator offset mismatches that affect the performance of conventional LU SAR architectures are detected by monitoring specific patterns in the non-binary SAR decisions without the need for dedicated analog detection hardware. This enables the design of the SubADC without the need for block reconfigurability, hence greatly reducing the calibration power overhead on analog blocks during normal operation.

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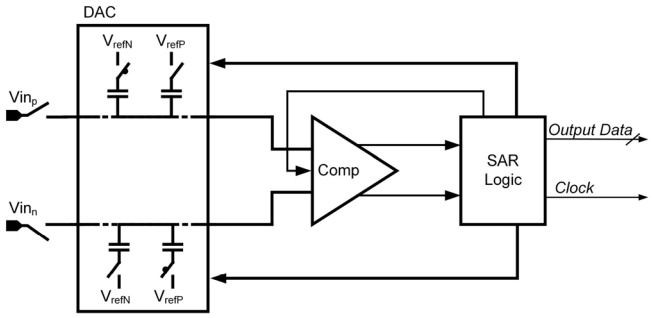


Fig. 1. Conventional SAR block diagram.

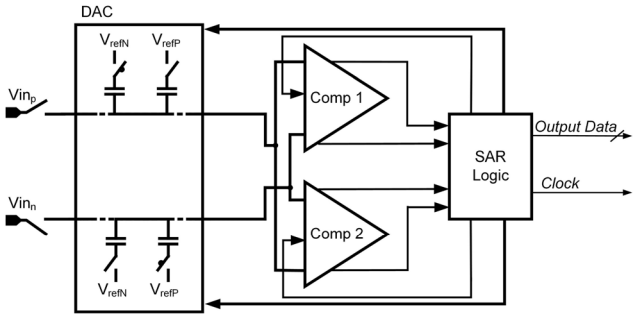


Fig. 2. Alternate comparator SAR block diagram.

This article is organized as follows. In Section II, different types of successive approximation architectures are reviewed and compared in terms of speed and noise-power efficiency. In Section III, the partial LU SAR scheme is described, and the calibration techniques used to detect comparator offset mismatches are then introduced in Section IV. The overall ADC architecture, time interleaving artifacts calibration, as well as the details of the implementation of the SubADC, sampling front end, and AFE are reported in Section V. Section VI presents the measurement results, and conclusions are drawn in Section VII.

II. Redundant Successive Approximation Architecture Efficiency Comparison

Successive approximation converters [12], [13] often leverage redundancy [14], [15], [16], in order to compensate for different sources of errors. This property has been leveraged to cope not only with digital to analog converter (DAC) settling time limitations [14] but also with DAC reference ripples [17] and every source of noise introduced during conversion. From an analog hardware perspective, we can introduce redundancy by properly sizing DAC weights w_n, such that for every cycle i, the sum of the remaining DAC weights is greater than w_i

$$\sum_{n=i+1}^{N_C} w_n > w_i \tag{1}$$

where N_c is the number of conversion steps. A possible way to introduce redundancy is to scale the DAC weights using an approximate geometric series with radix $\alpha < 2$. As an example, for a redundant converter resolving 7b in eight conversion steps, we could have

$$\{w_1, w_2, \ldots, w_8\} = \{34, 18, 10, 5, 4, 2, 1, 0.5\}. \tag{2}$$

Redundant successive approximation converters can be implemented using multiple architectures involving different speeds, complexity, and noise-power efficiency tradeoffs. A simplified comparison can be done by considering conversion time and power efficiency as metrics.

A. Conventional SAR Converter

In a conventional SAR converter (see Fig. 1), assuming that the DAC settling time is not a dominant contributor, the speed is limited by the time required for comparator decision and reset. The total conversion time t_{conv} could be thus written as

$$t_{\text{conv}} = N_c(t_{\text{dec}} + t_{\text{rst}}) \tag{3}$$

where t_{dec} and t_{rst} represent the comparator decision and reset time, respectively. During conversion, the main block contributing to noise is the comparator. This circuit is often implemented using fully dynamic topologies that draw energy (charge) from the supply only during the decision process. Assuming thermal noise as the dominant source, we can assume the following charge-noise relation:

$$Q_{\text{conv}} = E_0/v_n^2 \tag{4}$$

where E_0 is an efficiency coefficient lumping charge (Q_{conv}) and noise performance (v_n^2) of a given comparator topology for a reasonable sizing range.

For an N_c conversion steps SAR with a comparator noise power equal to v_{n0}, the total charge drawn during conversions is

$$Q_{\text{conv}}^{\text{SAR}} = N_c E_0/v_{n0}^2. \tag{5}$$

B. Alternate SAR Converter

A first improvement compared to a conventional SAR architecture is to duplicate the comparators and use them in an alternate scheme [18] (see Fig. 2). This scheme removes the need to wait for the comparator reset time as this operation is performed during the next bit decision. Neglecting as above the DAC settling time, the total conversion time can be written as

$$t_{\text{conv}} = N_c t_{\text{dec}}. \tag{6}$$

However, this solution does not allow optimization of comparator power compared to a conventional SAR. Both comparators are used during the last few LSB conversions, where no redundancy is available, so their noise will contribute to the overall converter noise performance. Furthermore, offset mismatches between the two comparators introduce conversion errors in the last decisions, hence requiring offset calibration [18], [19].

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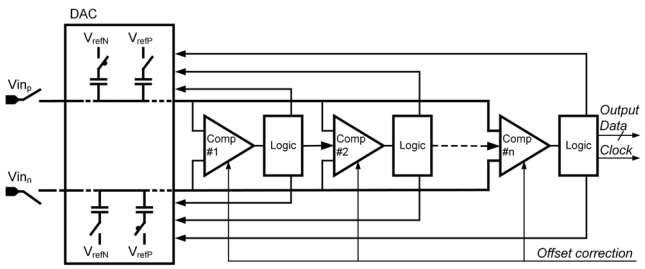


Fig. 3. LU SAR block diagram.

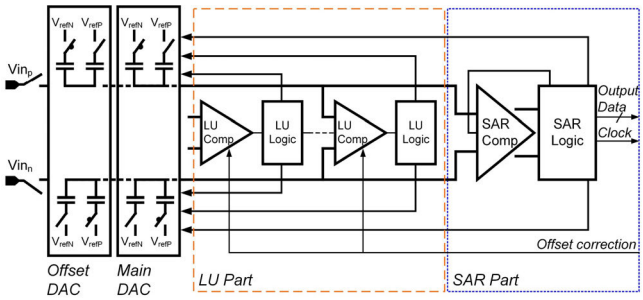


Fig. 4. Partial LU SAR block diagram and implementation details. The presented design features also an offset DAC and LU part consists of four coarse LU comparators.

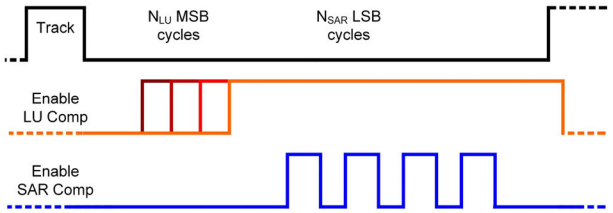


Fig. 5. Partial LU SAR timing diagram.

C. LU SAR Converter

An alternative with better intrinsic performance is the LU successive approximation architecture [20] (see Fig. 3). In this scheme, each decision is made using a dedicated comparator that can be properly sized for the optimal charge-noise tradeoff for the conversion steps where it is used. At the same time, speed is limited only by comparator decision times, as for the previous architecture. While it is possible to optimize each comparator sizing based on available redundancy ($\varepsilon_{\text{toli}}$), in practice, a limited number of sizing are used. In fact for MSB decisions, the optimum sized comparators may be smaller than minimum size devices available for a given technology. In addition to that, the amount of power saving due to this scaling yields diminishing returns as these blocks contribution to overall converter power becomes negligible compared to the ones used for LSB conversion steps. Furthermore, LSB decisions should still be made using comparators that have noise performance matching the target converter noise floor as for previous presented architectures. That is, for the same converter noise performance, comparators used in the LSB decisions of an LU architecture should have the same sizing as the ones used in conventional and alternate comparators SAR scheme.

A possible practical design choice for the purpose of this comparison is to use two comparators sizing (coarse and fine) that show 2× differences in rms noise and a resulting 4× difference in power.

Assuming N_{coarse} coarse and N_{fine} fine comparators are used, with $N_c = N_{\text{coarse}} + N_{\text{fine}}$, the total charge drawn from the supply can be written as

$$Q_{\text{conv}}^{\text{LU}} = N_{\text{coarse}} E_0/4v_{n0}^2 + N_{\text{fine}} E_0/v_{n0}^2. \tag{7}$$

If we choose, for example, $N_{\text{coarse}} = N_{\text{fine}}$, we have

$$Q_{\text{conv}}^{\text{LU}} = \left(\frac{N_c}{2} + \frac{N_c}{8}\right) E_0/v_{n0}^2 = \left(\frac{5}{8}\right) N_c E_0/v_{n0}^2 \tag{8}$$

which translates into a reduction of the power share of the comparators compared to conventional SAR by slightly less than 40%.

The downside of the LU architecture is that comparator offset mismatches will affect all the converter decision steps resulting in errors that may exceed redundancy. This is especially true during LSB decisions, where $\varepsilon_{\text{toli}}$ is either small or zero. The calibration of these mismatches is done in literature using an additional calibration phase [20]. This may involve additional conversion time, but most importantly, it requires reconfiguring converter building blocks to enable this operation. In fact, adding programmability may introduce power and speed penalties that affect the overall converter performance during normal operation.

III. Partial LU SAR SubADC Architecture

A possible solution to address the tradeoff of the above architectures is to merge the LU and the conventional SAR, yielding to a partial LU scheme (see Fig. 4). This concept has been introduced in [21], where the authors present a possible circuit-level implementation and simulation results.

The timing diagram is reported in Fig. 5. In this case, the first N_{LU} MSB steps are done using loop unrolling, leveraging dedicated loop unrolled comparator (LU Comp). The last N_{SAR} LSBs are performed using a conventional SAR featuring a single comparator reused at each conversion step.

In this scheme, we can reap the benefits of unrolling for the first N_{LU} conversion steps. At the same time, comparators can still be sized for the optimal charge-noise tradeoff.

For the simplified case of two comparators sizing adopted for the LU architecture, we can use the coarse one to implement the LU comparator and the fine one as SAR comparator. By further assuming $N_{\text{LU}} = N_{\text{SAR}}$, the total conversion time can thus be written as

$$t_{\text{conv}} = \frac{N_c}{2}(t_{\text{dec}} + t_{\text{rst}}) + \frac{N_c}{2} t_{\text{dec}} = N_c t_{\text{dec}} + \frac{N_c}{2} t_{\text{rst}}. \tag{9}$$

The total charge drawn from the supply is identical to the LU case, i.e.,

$$Q_{\text{conv}}^{\text{ParLU}} = \left(\frac{N_c}{2} + \frac{N_c}{8}\right) E_0/v_{n0}^2 = \left(\frac{5}{8}\right) N_c E_0/v_{n0}^2. \tag{10}$$

Table I reports the summary of the comparison among the architectures reviewed. In addition to conversion time and comparator charge per conversion, comparator area figure is

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TABLE I
ARCHITECTURES SIMPLIFIED CONVERSION TIME, CHARGE-NOISE EFFICIENCY, AND AREA COMPARISON

	SAR	Alternate SAR	Loop unrolled	Partial loop unrolled
Conversion time t_{conv}	$N_c t_{dec} + N_c t_{rst}$	$N_c t_{dec}$	$N_c t_{dec}$	$N_c t_{dec} + \frac{N_c}{2} t_{rst}$
Comparator charge per conversion Q_{conv}	$N_c E_0/v_{n0}^2$	$N_c E_0/v_{n0}^2$	$\left(\frac{5}{8}\right) N_c E_0/v_{n0}^2$	$\left(\frac{5}{8}\right) N_c E_0/v_{n0}^2$
Total comp. area (A_{coarse}=A_{fine}/4)	A_{fine}	$2A_{fine}$	$5A_{fine}$	$2A_{fine}$

reported on the very simplified assumption that comparator area scales with power ($A_{\text{coarse}} = A_{\text{fine}}/4$). Partial LU is a suitable compromise between complexity, speed, and charge-noise efficiency. Different tradeoffs are possible depending on the resolution, speed, and available comparator sizing, but this example can be used to highlight the underlying trend.

IV. SUBADC COMPARATOR OFFSET CALIBRATION

Partial LU architecture requires LU comparator offset mismatches to be calibrated to match the offset of the SAR comparator. This can be done using a calibration scheme that leverages the redundancy built in the DAC and the fact that the last steps are done as conventional SAR.

Indicating as V_{thk} the offset mismatch between the LU comparator used at conversion step k and the SAR one, two types of detection techniques can be used.

A. Dead Zone Detection

If $|V_{\text{thk}}| > \varepsilon_{\text{tolk}}$, the final quantization error may exceed 0.5 LSB. This will result in an error in the ADC transfer function, where the output code is not sensitive anymore to the input voltage. We denote this error a "dead zone." Referring again to the example of Section II, we can consider the DAC waveforms reported in Fig. 6, where the solid line represents a DAC transient waveform in the absence of errors, while the dashed one depicts a case where an error occurs at step 3. In this case, an offset $V_{\text{th3}} = 10$ LSB exceeds the available redundancy $\varepsilon_{\text{tol3}} = 3$ LSB and this results in a decision error. All following conversion steps are not affected by such errors and will try to compensate resulting in a string of decisions all equal to -1.

This error can be detected in the bit sequence for LU comparator used at step k as

$$\text{if } b_k = +1 \ \&\ b_i = -1 \quad \forall i > k \text{ then } V_{\text{thk}} > \varepsilon_{\text{tolk}} \tag{11}$$

$$\text{if } b_k = -1 \ \&\ b_i = +1 \quad \forall i > k \text{ then } V_{\text{thk}} < -\varepsilon_{\text{tolk}} \tag{12}$$

which allows to infer the sign of the offset V_{thk}.

With reference to DAC weights in (2) and the example depicted in Fig. 6, a sequence indicating a dead zone, that is an offset of LU comparator used in step 3 $V_{\text{th3}} > \varepsilon_{\text{tol3}}$, is simply

$$\{b_3, b_4, b_5, b_6, b_7, b_8\} = \{+1, -1, -1, -1, -1, -1\}.$$

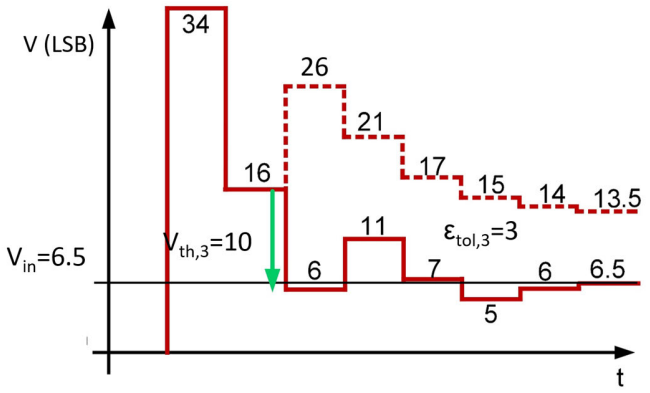


Fig. 6. LU comparator offset calibration DAC waveforms. Dead zone detection: no error (solid) and error > 0.5 LSB due to offset $V_{\text{th3}} > \varepsilon_{\text{tol3}}$ (dashed).

For the case $V_{\text{th3}} < -\varepsilon_{\text{tol3}}$, we have

$$\{b_3, b_4, b_5, b_6, b_7, b_8\} = \{-1, +1, +1, +1, +1, +1\}.$$

This detection strategy is limited as it stops when $-\varepsilon_{\text{tolk}} < V_{\text{thk}} < \varepsilon_{\text{tolk}}$. When this occurs, the transfer function of the converter will be linear, but all redundancy budget may be allocated to compensate for residual offset mismatch. This leaves the converter sensitive to any other error source. For example, noise from LU comparator may still affect converter performance.

B. Repeated Decision Detection

When $-\varepsilon_{\text{tolk}} < V_{\text{thk}} < \varepsilon_{\text{tolk}}$, no sequences of identical bit values are present; however, as per (1), it is possible to identify sequences, where the same DAC output $V_{\text{dac},k}$ is generated both at step k and at a different step j. If we consider, for example, the DAC transient waveform of Fig. 7, we can see that for some input signal ranges (e.g., $V_{\text{in}} = 15.3$), the same DAC value is generated at both steps $k = 3$ and $j = 8$. The former step is done during the LU phase, while the latter is performed during the SAR one. That means that, if we assume $V_{\text{th3}} = 0$, both b_3 (step 3) and b_8 (step 8) decisions will be identical. If instead $V_{\text{th3}} \neq 0$, b_k and b_j (b_3 and b_8 in this example) will not be equal. For the case where $V_{\text{th3}} = 1$ LSB in the previous example (see Fig. 8), b_3 decision will be affected by offset and will be equal to -1 but b_8 will be $+1$ as V_{th3} does not affect any longer the conversion. A similar case applies when $V_{\text{th3}} = -1$ LSB (see Fig. 9).

This condition can be detected in bit sequence as

$$\text{if } b_k = 1 \ \&\ b_j = -1 \ \&\ \sum_{i=k+1}^{j-1} b_i w_i = -b_k w_k \quad \text{then } V_{\text{thk}} > 0 \tag{13}$$

$$\text{if } b_k = -1 \ \&\ b_j = 1 \ \&\ \sum_{i=k+1}^{j-1} b_i w_i = -b_k w_k \quad \text{then } V_{\text{thk}} < 0. \tag{14}$$

In the actual implementation, both b_j and b_k decisions will be affected by thermal noise. In order to mitigate its impact,

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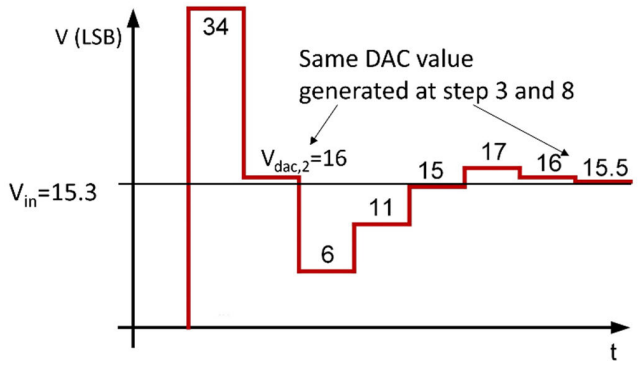


Fig. 7. LU comparator offset calibration DAC waveforms. Repeated decision detection: no offset error.

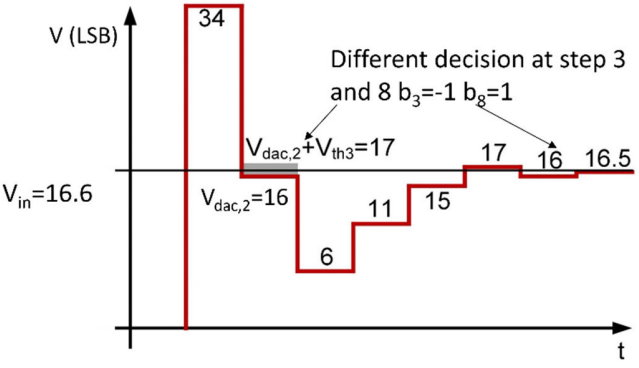


Fig. 8. LU comparator offset calibration DAC waveforms. Repeated decision detection: positive offset.

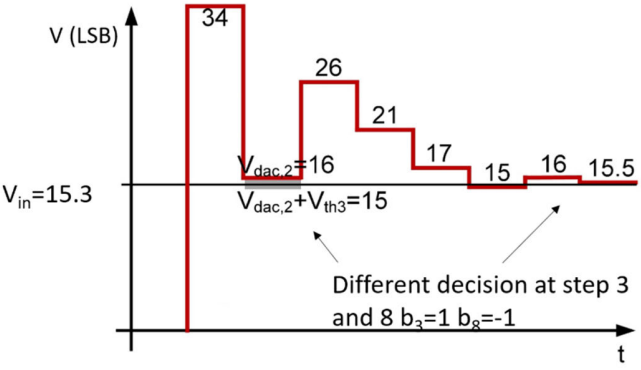


Fig. 9. LU comparator offset calibration DAC waveforms. Repeated decision detection: negative offset.

the information about V_{thk} offset can be extracted by averaging out multiple occurrences of such decisions when the conditions above are satisfied.

As example with reference to Figs. 7 and 8, the following sequences allow us to detect whether decisions b_3 and b_8 carry information about offset V_{th3}:

$$\{b_3, b_4, b_5, b_6, b_7\} = \{+1, -1, -1, -1, +1\}$$

and

$$\{b_3, b_4, b_5, b_6, b_7\} = \{-1, +1, +1, +1, -1\}.$$

When these sequences are detected, the DAC voltages generated at steps 3 and 8 are identical in fact with DAC weights

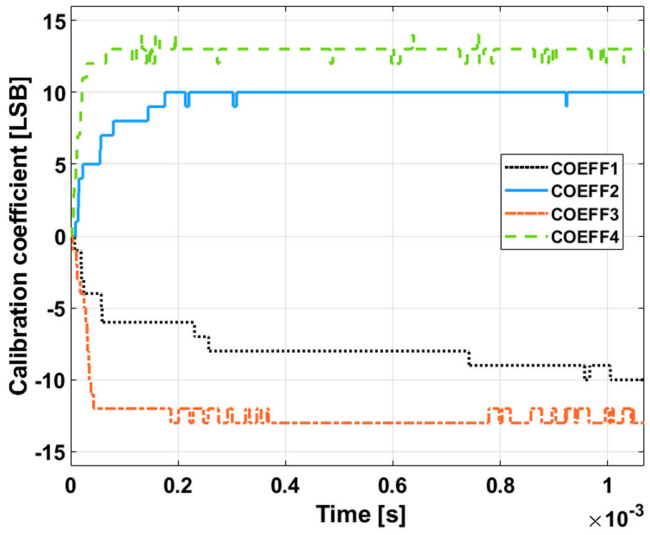


Fig. 10. Behavioral transient simulation of LU comparators offset calibration coefficient.

indicated in (2)

$$b_4 w_4 + b_5 w_5 + b_6 w_6 + b_7 w_7 = -b_3 w_3.$$

By detecting the proper bit sequences in digital, no dedicated analog hardware is required.

Compared to the calibration strategy presented in [21], the proposed approach enables full background tracking of coarse comparators offset drift over voltage and temperature. This is important to ensure good overall yield. In addition to that, no reconfigurability of the analog hardware or the timing critical SAR feedback path is required, hence improving conversion speed and power efficiency. Furthermore, no duplication of fine comparator is needed. On the other hand, as mentioned above, calibration conversion speed will be dependent on signal statistics.

The occurrence rate of the sequences to be detected depends on input signal statistics and this impacts calibration dynamic behavior and settling speed. For example, in order to get information about V_{th1}, the input signal should be close to the MSB decision threshold, which in a differential case is zero.

Similarly, to get information about V_{th2} (LU comparator used during MSB-1 decision), the signal to be converted should be reasonably close to the DAC voltage generated after MSB decision, that is, $\pm w_1$. Even assuming a uniform signal distribution in the amplitude domain, this results in different calibration settling times.

As an example, Fig. 10 shows a behavioral transient simulation of settling of calibration coefficients (ranging from -15 to 15) for a partial LU converter resolving 7b in eight conversion steps with $N_{\text{LU}} = 4$. In this case, the input signal amplitude probability density function has a uniform distribution between $-V_{\text{FS}}$ and $+V_{\text{FS}}$ (where V_{FS} is the ADC full scale). The calibration coefficient correcting for V_{th4} [coefficient 4 (COEFF4)] shows the fastest settling, while COEFF1 settling is the slowest as useful sequences are collected only close to a null differential input signal.

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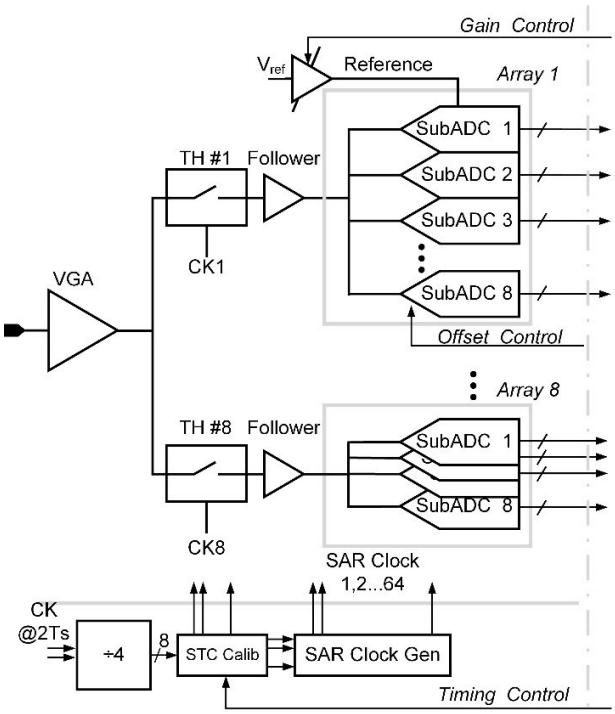


Fig. 11. AFE + ADC architecture.

In SerDes RX applications, input signal amplitude statistic distributions are usually adequate for proper calibration operation and resulting settling times are acceptable.

The coarse comparator offset values chosen in this example have been specifically chosen to highlight the different calibration settling speed behaviors. Extensive Monte Carlo behavioral simulations across coarse comparator offset values have been performed to ensure that no realistic offset combination may lead to calibration non-convergence. While this can be acceptable from a design engineering perspective, it may be topic for further theoretical investigation. Ripples around the final value are due to calibration operation and they do not affect the performance as they are small enough to be tolerated by available redundancy.

V. ADC + AFE Architecture and Implementation

The partial LU scheme and calibration presented have been used inside a TI ADC to get to the sampling speed required by the application.

The block diagram of the overall ADC is depicted in Fig. 11. It consists of a hierarchical TI array employing eight-way interleaved THs each driving eight SubADC channels using a buffer. The TH and SubADC are clocked by a clock generator block that derives the $8T_s$ and $64T_s$ sampling phases from a single half rate clock from PLL.

The ADC is driven by an AFE, which consists of a 100-Ω differential termination followed by a variable gain amplifier (VGA), which provides gain to adjust the input signal amplitude to match the ADC full-scale range.

A. Time Interleaving Calibrations

Time interleaving ADC converters require calibration of interleaving artifacts to achieve the target performance. In this design, TH sampling time skew is adjusted by the digital static timing calibration (STC) block operating on programmable delay lines.

A programmable reference generator is used for each of the eight SubADC array driven by the TH buffer. Gain mismatches are adjusted on a per TH base by changing the output voltage of the reference generator with an embedded DAC down to 10b equivalent resolution.

Finally, each SubADC includes a capacitive offset DAC to calibrate offset mismatches. The equivalent precision of the offset calibration is 8b, i.e., 1b more than the overall converter resolution.

During normal operation, these interleaving artifacts are calibrated in background by calculating suitable input signal statistics. While gain and offset information can be extracted also using sinewaves, STC block derives timing skew information by leveraging the properties of the application signal. From a general perspective, different techniques can be used to calibrate timing errors as reported in [22].

B. Partial LU SAR SubADC Implementation

The SubADC (see Fig. 4) adopts a partial LU architecture and resolves 7b in eight conversion steps. The system has been partitioned, such that the first four MSBs are done using loop unrolling, while the last four LSBs using SAR.

In this design, all LU comparators are equal and are optimized for maximum speed and minimum power. In this conversion phase, redundancy built in non-binary DAC mitigates the impact of their noise on overall SNR. A fine comparator is then used for the four LSB decisions. As in these last steps redundancy is small/absent, this comparator is optimized for maximum noise-power efficiency.

In this design, the coarse comparator is sized for 4× more input noise power compared to the fine one and draws 4× less input current.

The main DAC is implemented as a non-binary charge redistribution capacitive array. An auxiliary binary offset DAC sharing the same architecture is connected in parallel to calibrate 64 SubADC and eight TH offset mismatches.

LU comparators offset can be tuned using a resistive DAC (see Fig. 12) that degenerates each device of the input pair. This implementation ensures minimal power and area overhead as DAC elements have similar sizing as the input pair devices. When the offset is small (statistically the most likely case), all DAC transistors are ON ensuring maximum speed. The calibration algorithm selectively deactivates one side of the DAC pairs based on the sign of the offset. Degeneration of the input pair has a potential impact on input device transconductance (g_m) and thus on comparator noise performance. However, this additional degradation can be tolerated by the available redundancy.

The positive reference used for main and offset DACs is generated using two-stage slow op-amps with a PMOS output stage. The reference generator has a bandwidth significantly lower than the ADC conversion frequency and features internal decoupling that effectively filters high-frequency ripple. The positive reference value VrefP is around 700 mV, while the ground is used for VrefN.

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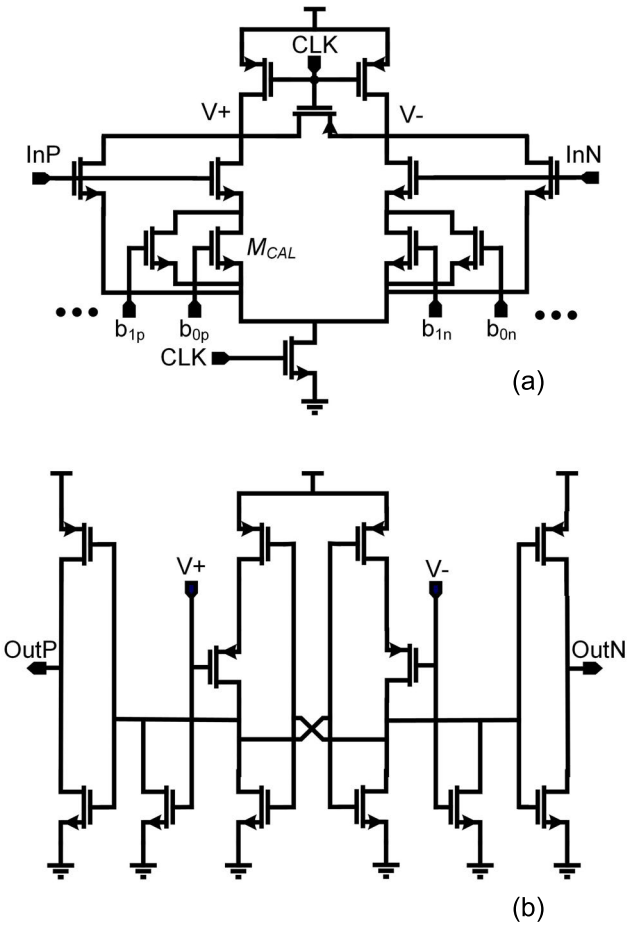


Fig. 12. Coarse comparator. (a) Input integrator. (b) Dynamic latch.

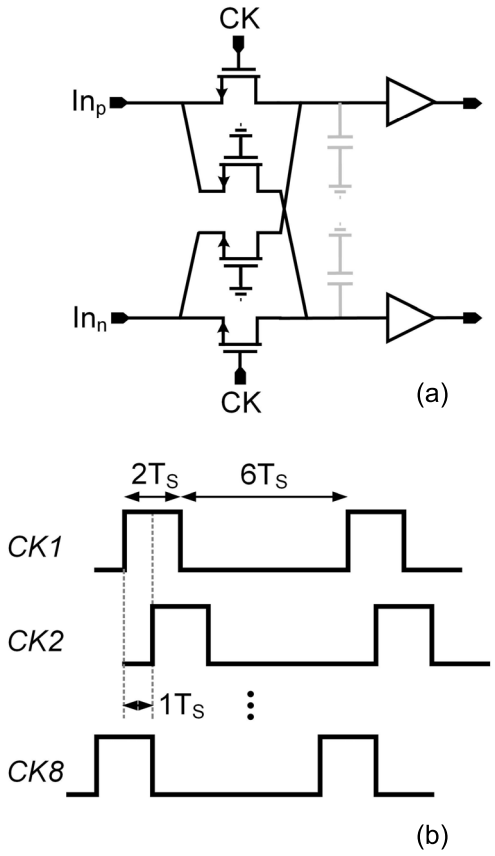


Fig. 13. TH. (a) Sampling switch. (b) Timing diagram.

C. Sampling Front End

The eight-way interleaved TH consists of an array of eight sampling units, each one composed of a sampling switch, followed by a buffer stage (see Fig. 13).

The NMOS sampling switch is driven by a CMOS logic control signal. As the input common mode voltage of the input signal is low (250 mV), there is no need for bootstrapped switches. The tracking time has a duration of $2T_s$, while the hold phase lasts for $6T_s$.

The overlapping of tracking time allows to achieve better linearity at the price of slightly larger input capacitance [23].

The choice of the sampling capacitance value is dictated by noise requirements. The capacitance is dominated by the input capacitance of the buffer stage. The input signal feedthrough caused by parasitic coupling during the hold phase is compensated by adding dummy cross-coupled devices between inputs and outputs of main sampling switches [2].

The sampling time mismatch of each sampling unit, resulting from device local random variations, can be corrected in the analog domain by operating eight independent delay lines. Each delay line is realized with a current-starved inverter, biased by a 7-bit low-speed current-steering DAC. Each inverter can delay the falling edge of the signal driving the sampling switch gate, thus controlling the sampling instant. The timing resolution of the delay line is 100 fs. The non-linearity of the transfer function of delay versus input

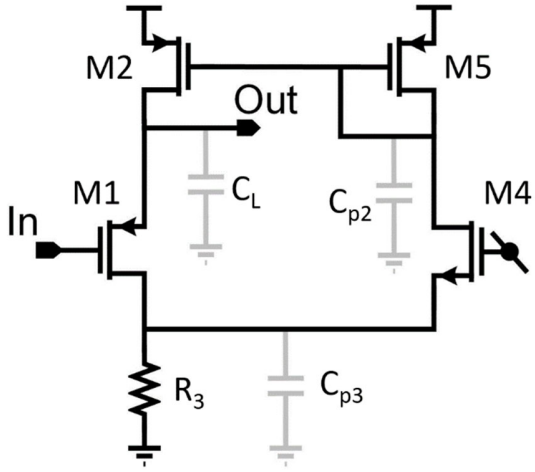


Fig. 14. TH buffer implementation.

DAC code is not a concern as the delay lines are closed in the calibration loop. The DSP adjusts the delay line control code until the error stemming from timing mismatches is minimized. A current-controlled topology has been preferred to a shunt capacitance one as the former shows a better jitter/range/power tradeoff [24].

The buffer following the sampling switch (see Fig. 14) has $6T_s$ to charge the SAR sampling capacitance and the superfollower architecture enhances its output charging current capability [25].

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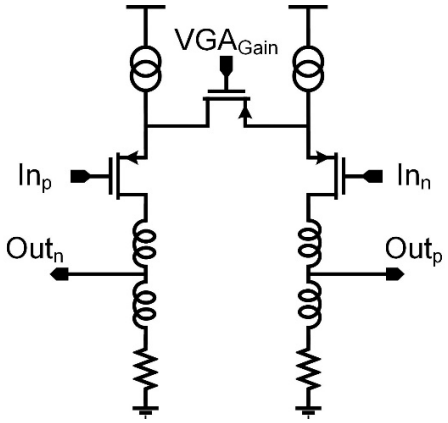


Fig. 15. VGA implementation.

The PMOS input stage ensures that the common mode voltage provided to the following SAR comparators is at an optimum point for noise and speed performance. The buffer is biased with a resistance to reduce the amount of parasitic capacitance in the loop and improve stability.

When the feedback loop is opened for stability analysis, the resulting transfer function consists of three poles, whose approximate analytical expressions are reported as follows:

$$f_{p1} \cong \frac{g_{m1}}{2\pi C_L} \quad f_{p2} \cong \frac{g_{m5}}{2\pi C_{p2}} \quad f_{p3} \cong \frac{g_{m4} + R_3^{-1}}{2\pi C_{p3}}. \tag{15}$$

The resulting approximated dc value of the open-loop gain G_{loop} can be written as

$$G_{\text{loop}} \cong \frac{g_{m2}}{g_{m5}} \frac{g_{m4}}{R_3^{-1} + g_{m4}} \leq \frac{g_{m2}}{g_{m5}}. \tag{16}$$

The circuit is sized, such that the dominant pole is the one related to output load capacitance C_L (f_{p1}). The dc value of G_{loop} is set to few decibels in order to enable a more symmetrical current sourcing and sinking capability during large transient signal settling compared to simple source followers.

D. Variable Gain Amplifier (VGA)

The VGA (see Fig. 15) used in the AFE consists of a single-stage PMOS-based current mode logic (CML) amplifier using programmable MOS as resistive degeneration. Bandwidth is extended by inductive techniques beyond 30 GHz. The maximum dc gain is approximately 6 dB [26]. For the target application, no analog peaking is required as equalization will be mostly done in the digital domain.

The VGA drives two THs in tracking state plus six THs in hold state. The intersymbol interference (ISI) coming from the previously sampled values at $7T_S$ and $8T_s$ in the past is minimized by properly sizing the driving strength of the VGA. Moreover, a feed-forward equalizer (FFE) [27] in the digital domain can compensate for this memory effect.

VI. Measurements

The prototype is fabricated in 5-nm FinFET technology, and Fig. 16 shows the die photograph. The total area is 0.151 mm^2. The measurements are done with the converter operating at 60 GS/s and with VGA configured for maximum gain. As the VGA cannot be bypassed, all measurements include its distortion, power, and noise contribution.

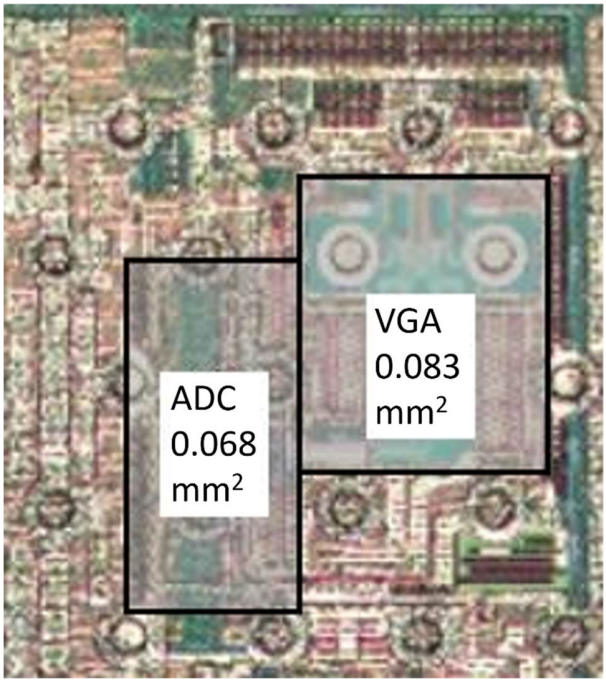


Fig. 16. Die photograph.

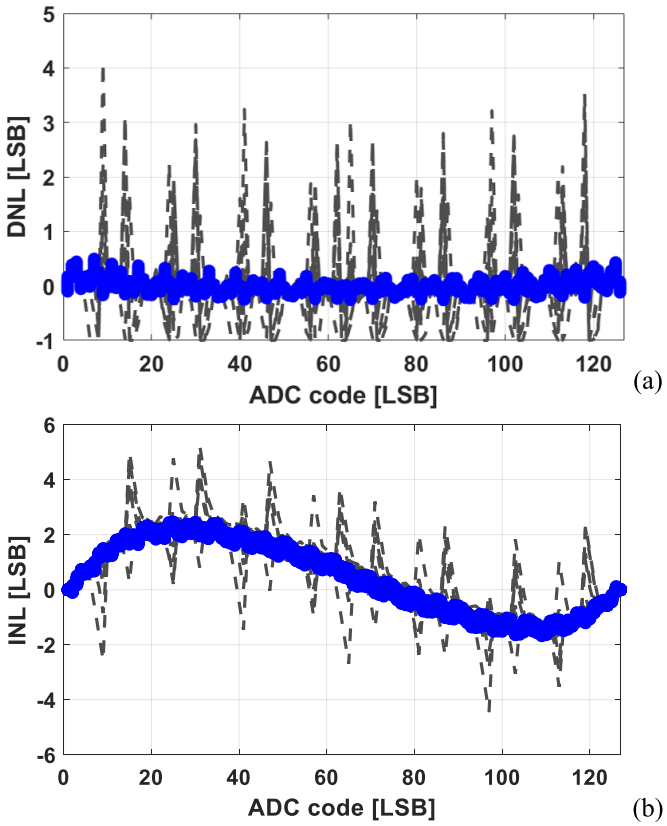


Fig. 17. (a) DNL and (b) INL overlay of 64 SubADC before (dashed gray) and after (solid blue) comparator offset calibration.

Fig. 17(a) reports the differential non linearity (DNL) of all 64 SubADCs before and after comparator offset calibration.

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TABLE II
PERFORMANCE COMPARISON

	This work	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
Technology	5nm	14nm	7nm	7nm	5nm	4nm	16nm	5nm	22nm	16nm
Fs (GS/s)	60	72	30	97	112	56	105	200	40	42
Resolution (bits)	7b	8b	7b	8b	6b	8b	8b	8b	7b	7b
Power (mW)	109.3	235	53.5	311	189	240	698	400	66.1	96
SNDR @ LF (dB)	35.3	39.3	NA	44.5	NA	41.0	42.8	41.5	37.3	32.5
SNDR @ HF (dB)	35.5/ 35.2	34.2	32.0	35.1	28.9	32.8	39.2	36.1*	32.3	28.9
FoMw @ HF (fJ/c.s.)	37.7/ 38.75	77.91	54.84	67.00	74.63	120.2	89.22	38.36*	49.09	100.4
HF used for comparison (GHz)	20/32	20	13.5	30	56	26.9	20.9	20	20	20.8
AFE included?	yes (Power and SNDR)	no	yes (SNDR only)	yes (Power and SNDR)	yes (SNDR only)	No	yes (SNDR only)	yes (Power and SNDR)	yes (Power and SNDR)	yes (Power and SNDR)
Notes					SNDR from Digest [21]			*Extrap. at 32G: 34.7dB and 45fJ/c.s.		No slope gen.

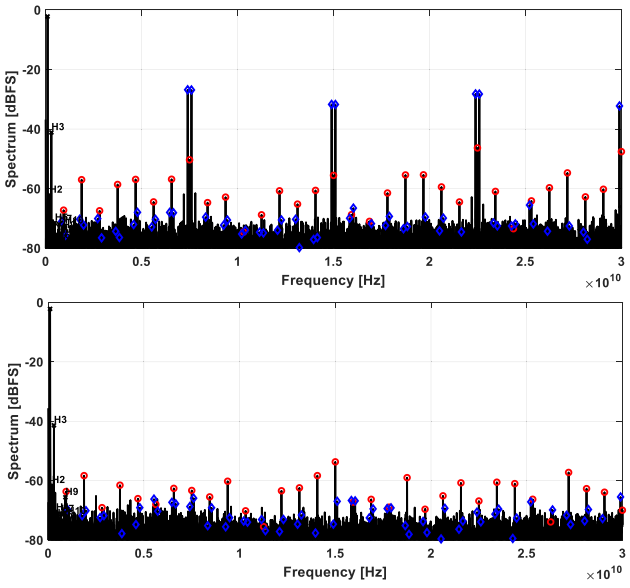


Fig. 18. Output spectrum for low frequency input before (top) and after (bottom) time interleaving artifacts calibration.

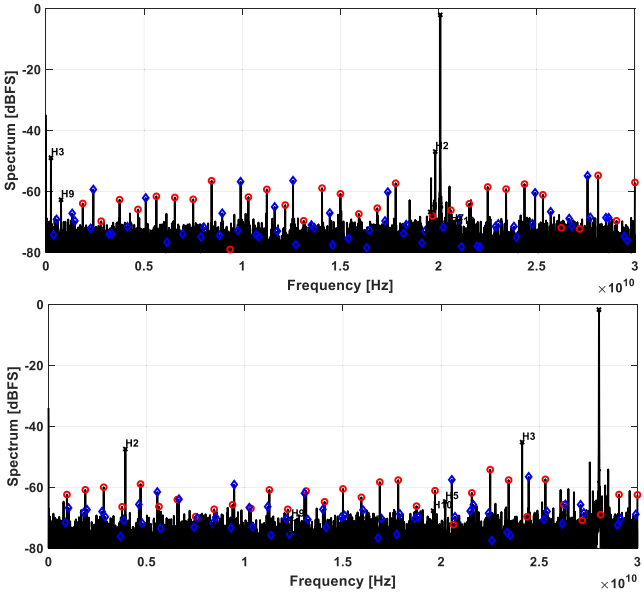


Fig. 19. Output spectrum for 20-GHz (top) and 32-GHz (bottom) input.

Missing codes in the precalibration curves appear when the input signal amplitude values are such that LU comparator offset affects the decision, generating dead zones. By activating the LU comparators calibration block, missing codes are effectively removed from the transfer function and DNL improves to 0.5/−0.2 LSB. INL curves of 64 subADCs before and after calibration are depicted in Fig. 17(b). The residual sine shape like trend in INL after calibration could be ascribed to VGA non-linearities as the curves have been derived using the sinewave fitting histogram method.

A spectrum with a $-0.7\text{dB}_{\text{FS}}$ (full-scale $0.5V_{\text{ppd}}$) low-frequency input signal is shown in Fig. 18 before and after calibration of time interleaving artifacts. Large gain mismatch tones before calibration can be mainly ascribed to mismatches among the eight reference generators. In these measurements, gain and offset calibrations are operating in background. Timing skew errors are calibrated in foreground using the actual application signal, and then, calibration coefficients are kept constant during sinewave measurements.

After calibration of interleaving artifacts with a 20 and 32-GHz input, the signal to noise and distortion ratio (SNDR) of ADC + VGA is 35.5 and 35.2 dB, respectively (see Fig. 19).

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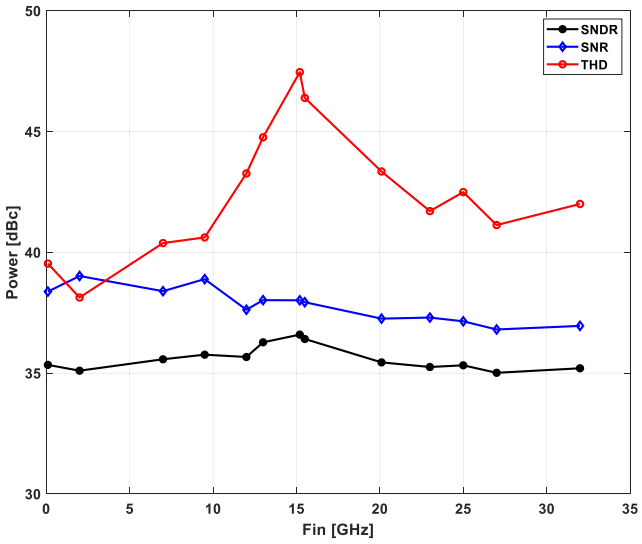


Fig. 20. SNR, SNDR, and THD versus input signal frequency.

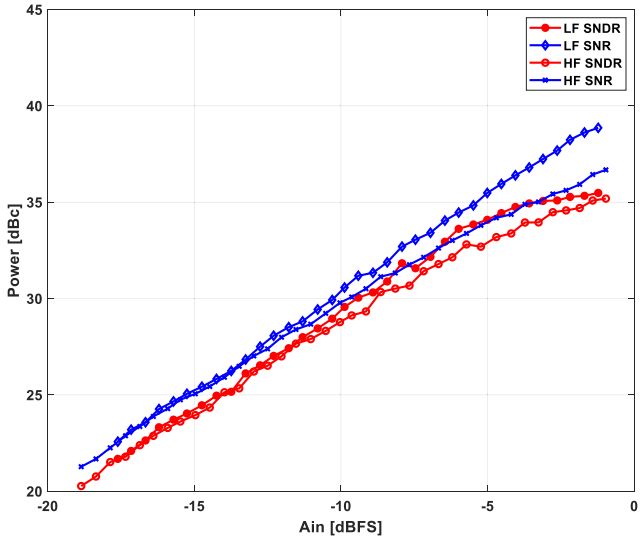


Fig. 21. SNR and SNDR versus input signal amplitude.

This figure is improved compared to previously reported results [28] due to the use of a better bench signal source and overall measurement setup.

SNDR, SNR, and THD versus input signal frequency are reported in Fig. 20. SNR roll off with input signal frequency can be mainly ascribed to wideband jitter. The jitter extrapolated from the measured data is approximately 60fs$_{rms}$.

Performance as a function of the input signal amplitude, for low- and high-frequency inputs, is shown in Fig. 21.

At 60 GS/s, the ADC + VGA including calibration draws 109.3 mW from a single 0.9 V supply. This figure includes all auxiliary blocks as well as LU comparator calibration logic. The Walden figure of merit (FoM) of the presented work with 20- and 32-GHz inputs are 37.7 and 38.75 fJ/c.s., respectively.

A detailed comparison with recently published ADCs in the same speed/SNDR range (see Table II) [29] shows that this work achieves the state-of-the-art power efficiency for FinFET ADCs at sampling rates beyond 30-GS/s reporting performance beyond Nyquist frequency [2], [3], [4], [5], [6], [7], [8], [9], [10].

VII. Conclusion

In this article, we described a 7b 60-GS/s 64-way interleaved ADC with integrated VGA AFE that delivers 35.2 dB of SNDR till 32-GHz input frequency, that is, beyond Nyquist. The total power consumption is 109.3 mW, including AFE, LU comparator calibration logic, and all auxiliary blocks. The prototype fabricated in 5-nm FinFet technology shows a total area of 0.151 mm^2. The key enabling technology for this design is the usage of a partial LU SAR SubADC architecture featuring an LU comparator calibration technique that does not require any analog hardware overhead. It achieves a Walden FoM of 37.7 and 38.75 fJ/c.s. at 20- and 32-GHz input frequency, respectively, which compares favorably with respect to state of the art.

Acknowledgment

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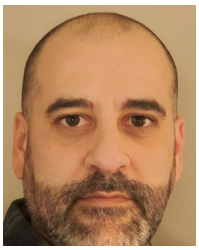
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From 2007 to 2011, he was an Analog IC Designer involved in the development of NAND flash memories. From 2011 and 2018, he joined STMicroelectronics, as an RFIC Analog Designer, developing ICs in SiGe BiCMOS for wireless infrastructure market and then for Fiber Optics and Silicon Photonics applications. From 2018 to 2021, he was with Huawei, Segrate, Italy, developing Optical RFICs in SiGe BiCMOS for optical communications. Since 2021, he has been working as an Analog IC Designer with Marvell Technology, Pavia, Italy, on FinFET high-speed electro-optical interfaces, with particular interest for TIAs.



Nicola Codega was born in Sondrio, Italy, in 1986. He received the M.Sc. degree (summa cum laude) and the Ph.D. degree in electronic engineering from the University of Pavia, Pavia, Italy, in 2010 and 2013, respectively.

He was the Ph.D. intern in Marvell, Pavia, from 2010 to 2013, and Marvell, Aliso Viejo, CA, USA, in 2012, working on multistandard reconfigurable transceivers for mobile communications. He has been working as an Analog Design Engineer, since 2013, and in 2021, he joined Marvell Technology Group, Pavia, through Inphi Corp. acquisition. He is currently working on transceivers for high-speed serial communications.



Alessio Di Pasquo was born in Rome, Italy, in 1995. He received the B.S., M.S., and Ph.D. degrees in electronic engineering from the University of Pisa, Pisa, Italy, in 2017, 2019, and 2022, respectively.

From 2019 to 2021, he was the Ph.D. intern at Inphi Corp, Pavia, Italy. Following Inphi Corp's acquisition by Marvell Technology Group, in 2021, he continued his internship, focusing on time-interleaved track and hold circuits for serializer/deserializer (SerDes) applications. Since 2022, he has been working as an Analog Design Engineer. He is currently involved in developing transceivers for high-speed serial communications, with a particular emphasis on receiver front ends and analog-to-digital converters.



Ivan Fabiano was born in Milan, Italy, in 1986. He received the B.S. and M.S. degrees in electrical engineering and the Ph.D. degree with a focus on integrated analog system for mobile applications from the University of Pavia, Pavia, Italy, in 2009, 2011, and 2014, respectively.

From 2014 to 2017, he has been with Marvell Semiconductor Inc., where he was involved in the high speed data interfaces development focusing on receiver (RX) AFE. In 2017, he joined eSilicon, Pavia, where he was developing transmitter (TX) and RX for long-reach electrical serializer/deserializer (SerDes). In 2020, he joined Inphi Corp., Italy, through an acquisition, where he focused on optical transceivers. In 2021, he joined Marvell Technology Group through Inphi Corp. acquisition, where is still focused on high-speed optical transceivers. His current research interests include high-speed mixed signal, radio frequency circuits and systems, and high-speed serial link.



Marco Garampazzi was born in Milan, Italy, in 1986. He received the M.Sc. degree (summa cum laude) and the Ph.D. degree in electronic engineering from the University of Pavia, Pavia, Italy, in 2011 and 2014, respectively.

During the Ph.D., he interned at Marvell Semiconductor, Aliso Viejo, CA, USA. Since 2014, he has been working as an Analog Design Engineer, focusing on RF front ends for multistandard transceivers. In 2021, he joined Marvell Technology Group through the acquisition of Inphi Corp., where he holds a senior design role. Currently, he is working on phase-locked loop circuits and low-phase-noise oscillators for high-speed communications.



Fabio Giunco (Member, IEEE) was born in Tortona, Italy, in 1971. He received the M.S. degree in electronic engineering from the University of Pavia, Pavia, Italy, in 1996.

Since then, he has worked in the field of analog and mixed signal design. He began as fresh graduated at Accent srl 1997–2007, where he developed chips for a variety of applications, including low-noise sensor interface for automotive, power management, and low-power wireless. He was a design team Lead at STMicroelectronics, Milan, Italy, 2007–2014, where he worked in the HD read–write channel leading the analog-to-digital converter (ADC) team and later developing serializer/deserializer (SerDes) for PCIexpress. He currently works as a Design Director and Project Manager with Marvell Technology, Pavia, where he has contributed to the development of Spica optical modules. His current research interests include high-speed interfaces and optical interfaces.



Leonardo Daniel Herbas Burgos was born in Ecuador, in 1990. He received the M.Sc. degree in microelectronics engineering from the University of Pavia, Pavia, Italy, in 2020.

He was an intern in eSilicon, Pavia, during the master's as part of the project LM+ with focus on octave tuning range voltage controlled oscillators (VCOs). He has been working as an Analog Design Engineer at Marvell Technology Group through Inphi Corp. acquisition, since 2020. He is currently working on high-speed circuits for optical links.

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Gabriele Minoia received the B.S. and M.S. degrees in electronic engineering from the University of Pavia, Pavia, Italy, in 2004 and 2006, respectively.

From 2007 to 2010, he held a grant from the University of Pavia within Studio di Microelettronica, STMicroelectronics, Catania, Italy, working on the design of gm-C filters for hard disk drive (HDD) read and write (R/W) channel applications. From 2010 to 2019, he has been with STMicroelectronics, working on the development of high-speed ICs for silicon photonics applications in CMOS and BiCMOS technologies. In 2019, he joined eSilicon, Pavia, working on the development of die-to-die parallel interfaces. He joined Marvell, Pavia, in 2021, through eSilicon/Inphi acquisition; he is now a Principal Analog Design Engineer, working on different aspects of high-speed interfaces and frequency synthesis.

Paolo Rossi, photograph and biography not available at the time of publication.



Marco Sosio was born in Tirano, Italy, in 1985. He received the M.S. degree (summa cum laude) in electronics engineering and the Ph.D. degree in microelectronics from the University of Pavia, Italy, in 2009 and 2012, respectively.

During Ph.D., he was intern at Marvell, Pavia and Ericsson AB, Lund, Sweden. Research activity had focus on filtering analog-to-digital converter (ADC)-based baseband for wireless applications. From 2012 to 2017, he was in Marvell Semiconductor Analog Signal Processing and Serializer/Deserializer (SerDes) team, developing analog baseband, SAR analog-to-digital converter (ADC), and digital to analog converter (DAC) for audio–video near field communication (NFC) applications and high-speed serial links. In 2017 and 2020, he joined eSilicon, Pavia, and Inphi, respectively, where he owned transmitter (TX) activities for high-speed wireline communications (IP and pluggable modules market). In 2021, he joined again Marvell Technology, Pavia, working on TXs for serial link and co-owning analog design of transceivers for data-center high-speed communications.



Leonardo Vignoli was born in Florence, Italy, in 1996. He received the B.S. and M.S. degrees in electronics engineering from the University of Florence, Florence, in 2019 and 2022, respectively.

During the master's thesis, he joined Marvell Technology, Pavia, Italy, as an intern, where he conducted research on the impact of power supply noise and disturbances on the performances of serializer/deserializer (SerDes) circuits, where he is currently a Post-Silicon Validation Engineer, focusing on the analog–digital characterization and standard compliance of high-speed optoelectrical serial interfaces.



Enrico Temporiti received the Laurea degree in electronic engineering from the University of Pavia, Pavia, Italy, in 1999, in collaboration with Alcatel Italia.

In 2000, he joined STMicroelectronics, Studio di Microelettronica, Pavia, where he held various roles across multiple organizations, ranging from advanced research and development to strategic planning and product development. Since 2019, he has been a Director of Engineering at Marvell Technology, Pavia (formerly Inphi Italy and eSilicon Italy), leading the development of electrical and optical physical tranceiver (PHYs). He holds several U.S. and European patents, primarily in the fields of optical communications and frequency synthesis. His technical expertise includes CMOS analog and mixed-signal high-speed integrated circuits for wireless and wireline communications, with a focus on RF frequency synthesizers, silicon photonics transceivers, and chiplet die-to-die interfaces.



Shawn Scouten received the B.Sc. degree engineering physics/electrical from Queen's University, Kingston, ON, Canada, in 1991, and the M.S. degree in applied physics/electrical from Cornell University, Ithaca, NY, USA, in 1993.

From 1993 to 2001, he worked at Nortel Network, Ottawa, ON, Canada, in analog design with a focus on optical transport serializer/deserializer (SerDes). Since 2001, he has been in a Lead role with an analog design team that has evolved through Cortina Systems, Inphi, and Marvell, working to deliver more than 30 non return to zero (NRZ)/pulse amplitude modulation (PAM) optical and wireline CMOS SerDes products to volume production with a focus on data center applications.



Stephen Jantzi received the M.A.Sc. and Ph.D. degrees from the University of Toronto, Toronto, ON, Canada, in 1992 and 1997, respectively, where he worked on the first silicon implementations of bandpass and quadrature bandpass delta–sigma analog-to-digital converters (ADCs).

From 1997 to 2005, he was with Broadcom, Irvine, CA, USA, where he was an analog Lead for multiple generations of cable modem and cable-set-top box chips. From 2005 to 2013, he was with Fresco Microchip and Spectra7 Microsystems leading silicon TV tuner and demodulator chips. He joined Clariphy communications in 2014 to lead the analog team working on high-speed coherent optical communications. Since the acquisition of Clariphy by Inphi in 2016 and the subsequent acquisition of Inphi by Marvell in 2021, he has continued leading the Inphi and Marvell analog teams for high-speed coherent and optical-pulse amplitude modulation (PAM) connectivity devices. He is currently the Vice President of Engineering of Marvell's optical PHY team.

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